



# KG-2 BOOK B



Decomposition of 10 & Big/Small Friends

## Student Information

Name:

.....

Level:

.....

Date:

.....

8

9

5



# Combination of Big & Small Friends

Teen – single rewrites as – 10 + (10 – single)



★

$$13 - 8 : 13 - 10 + 2 = ( )$$



✱

$$\begin{array}{r} 13 \\ - 8 \\ \hline \end{array}$$

★

☰

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$$14 - 9 : 14 - 10 + 1 = ( )$$



$$\begin{array}{r} 14 \\ - 9 \\ \hline \end{array}$$



# Decomposition Pyramid

 Solve every subtraction problem in the staircase



|      |      |      |      |      |      |      |      |      |  |
|------|------|------|------|------|------|------|------|------|--|
| 10-1 |      |      |      |      |      |      |      |      |  |
| 10-2 | 11-2 |      |      |      |      |      |      |      |  |
| 10-3 | 11-3 | 12-3 |      |      |      |      |      |      |  |
| 10-4 | 11-4 | 12-4 | 13-4 |      |      |      |      |      |  |
| 10-5 | 11-5 | 12-5 | 13-5 | 14-5 |      |      |      |      |  |
| 10-6 | 11-6 | 12-6 | 13-6 | 14-6 | 15-6 |      |      |      |  |
| 10-7 | 11-7 | 12-7 | 13-7 | 14-7 | 15-7 | 16-7 |      |      |  |
| 10-8 | 11-8 | 12-8 | 13-8 | 14-8 | 15-8 | 16-8 | 17-8 |      |  |
| 10-9 | 11-9 | 12-9 | 13-9 | 14-9 | 15-9 | 16-9 | 17-9 | 18-9 |  |



# Abacus Calculation: Decomposition of 10 (–9)



–9 = –10 + 1    Fingering Ex.: 10–9, 11–9, 12–9

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 10 | 11 | 12 | 13 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 2  | 3  | 4  | 1  | 5  | 2  | 3  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 14 | 15 | 16 | 17 | 18 | 11 | 12 | 13 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 3  | 2  | 1  | 3  | 2  | 4  | 5  | 6  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 14 | 15 | 16 | 17 |
| 2     | -9 | -9 | -9 | -9 |
| 3     | 2  | 3  | 4  | 1  |
| Ans.  |    |    |    |    |



11 – 9 =



12 – 9 =



13 – 9 =

13 – 9

= 13-10+1

= 4



# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 | 16 | 17 | 10 | 11 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 3  | 5  | 2  | 4  | 3  | 2  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 12 | 13 | 14 | 15 | 16 | 10 | 11 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 1  | 5  | 2  | 3  | 4  | 2  | 3  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | ☆ |
|-------|----|----|----|----|---|
| 1     | 12 | 13 | 14 | 15 |   |
| 2     | -9 | -9 | -9 | -9 |   |
| 3     | 4  | 3  | 2  | 1  |   |
| Ans.  |    |    |    |    |   |



$11 - 8 =$



$13 - 7 =$



$15 - 6 =$



$12 - 9 =$

# Abacus Calculation: Decomposition of 10 (–8)



–8 = –10 + 2    Fingering Ex.: 11–8, 13–8, 15–8, 16–8

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 2     | -8 | -8 | -8 | -8 | -8 | -8 | -8 | -8 |
| 3     | 3  | 4  | 2  | 5  | 3  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 | 11 | 13 | 15 | 17 |
| 2     | -8 | -8 | -8 | -8 | -8 | -8 | -8 | -8 |
| 3     | 5  | 3  | 2  | 4  | 6  | 1  | 5  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 |
| 2     | -8 | -8 | -8 | -8 |
| 3     | 4  | 2  | 3  | 1  |
| Ans.  |    |    |    |    |

11 – 8 =

13 – 8 =

14 – 8 =

11 – 8  
= 11-10+2  
= 3

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 10 | 11 | 14 | 16 | 15 | 13 | 12 |
| 2     | -9 | -8 | -9 | -8 | -9 | -8 | -9 | -8 |
| 3     | 3  | 4  | 2  | 3  | 4  | 2  | 3  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 |
| 2     | -9 | -8 | -9 | -8 | -9 | -8 | -9 | -8 |
| 3     | 2  | 3  | 4  | 1  | 5  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 18 | 16 | 14 | 12 |
| 2     | -9 | -8 | -9 | -8 |
| 3     | 3  | 1  | 4  | 2  |
| Ans.  |    |    |    |    |

$15 - 7 =$

$14 - 6 =$

$11 - 5 =$

$12 - 4 =$

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 2     | -8 | -9 | -8 | -9 | -8 | -9 | -8 | -9 |
| 3     | 3  | 2  | 4  | 1  | 3  | 4  | 2  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 2     | -9 | -8 | -9 | -8 | -9 | -8 | -9 | -8 |
| 3     | 4  | 5  | 3  | 2  | 4  | 1  | 3  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 |
| 2     | -8 | -9 | -8 | -9 |
| 3     | 3  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |

13 – 4 =

16 – 9 =

15 – 8 =

14 – 7 =



# Abacus Calculation: Decomposition of 10 (-7)



-7 = -10 + 3    Fingering Ex.: 11-7, 15-7, 16-7

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 4  | 3  | 2  | 5  | 3  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 | 11 | 13 | 15 | 17 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 5  | 3  | 2  | 4  | 6  | 1  | 5  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 |
| 2     | -7 | -7 | -7 | -7 |
| 3     | 4  | 2  | 3  | 1  |
| Ans.  |    |    |    |    |

11 - 7 =

13 - 7 =

15 - 7 =

11 - 7  
= 11-10+3  
= 4

# Visual Subtraction

Subtract 6 from each — what is left?



$$11 - 6 = \boxed{\phantom{00}}$$



$$13 - 6 = \boxed{\phantom{00}}$$



$$16 - 6 = \boxed{\phantom{00}}$$



$$15 - 6 = \boxed{\phantom{00}}$$

# Visual Subtraction

Different starting numbers, same trick



$$10 - 6 : 10 - 10 + 4 = ( )$$



$$\begin{array}{r} 10 \\ - 6 \\ \hline \end{array}$$

$$15 - 6 : 15 - 10 + 4 = ( )$$



$$\begin{array}{r} 15 \\ - 6 \\ \hline \end{array}$$



# Abacus Calculation: Decomposition of 10 (–6)



–6 = –10 + 4    Fingering Ex.: 10–6, 15–6, 16–6

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 2     | -6 | -6 | -6 | -6 | -6 | -6 | -6 | -6 |
| 3     | 2  | 4  | 3  | 5  | 1  | 3  | 2  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 12 | 13 | 14 | 15 | 16 | 10 | 11 |
| 2     | -6 | -6 | -6 | -6 | -6 | -6 | -6 | -6 |
| 3     | 5  | 3  | 2  | 5  | 1  | 3  | 4  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 |
| 2     | -6 | -6 | -6 | -6 |
| 3     | 5  | 4  | 2  | 3  |
| Ans.  |    |    |    |    |

10 – 6 =

12 – 6 =

14 – 6 =

15 – 6  
= 15-10+4  
= 9

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 15 | 16 | 17 | 18 | 14 | 13 | 12 | 11 |
| 2     | -6 | -7 | -8 | -9 | -6 | -7 | -8 | -9 |
| 3     | 3  | 4  | 2  | 1  | 5  | 3  | 4  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 2     | -6 | -7 | -8 | -9 | -6 | -7 | -8 | -9 |
| 3     | 4  | 3  | 2  | 1  | 5  | 3  | 4  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 18 | 16 | 14 | 12 |
| 2     | -6 | -7 | -8 | -9 |
| 3     | 5  | 3  | 4  | 2  |
| Ans.  |    |    |    |    |

10 – 5 =

12 – 5 =

11 – 4 =

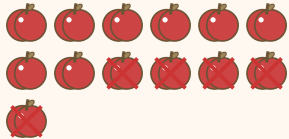
13 – 3 =

# Visual Subtraction Cards

Three more subtraction tricks



$$13 - 5 : 13 - 10 + 5$$



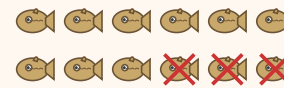
$$\begin{array}{r} 13 \\ - 5 \\ \hline \end{array}$$

$$11 - 4 : 11 - 10 + 6$$



$$\begin{array}{r} 11 \\ - 4 \\ \hline \end{array}$$

$$12 - 3 : 12 - 10 + 7$$



$$\begin{array}{r} 12 \\ - 3 \\ \hline \end{array}$$



# Abacus Calculation: Decomposition of 10 (-5)



-5 = -10 + 5    Fingering Ex.: 10-5, 11-5, 12-5

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 2     | -5 | -5 | -5 | -5 | -5 | -5 | -5 | -5 |
| 3     | 3  | 2  | 4  | 1  | 5  | 3  | 2  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 2     | -5 | -5 | -5 | -5 | -5 | -5 | -5 | -5 |
| 3     | 3  | 5  | 4  | 3  | 2  | 1  | 5  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 |
| 2     | -5 | -5 | -5 | -5 |
| 3     | 4  | 2  | 3  | 1  |
| Ans.  |    |    |    |    |

$10 - 5 =$

$11 - 5 =$

$12 - 5 =$

$13 - 5$   
 $= 13 - 10 + 5$   
 $= 8$

## Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------|----|----|----|----|----|----|----|----|----|----|
| 1     | 11 | 31 | 20 | 43 | 14 | 35 | 26 | 24 | 41 | 12 |
| 2     | -5 | -9 | -4 | -8 | -6 | -7 | -8 | -5 | -9 | -4 |
| 3     | 2  | 4  | 3  | 5  | 2  | 1  | 3  | 4  | 2  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------|----|----|----|----|----|----|----|----|----|----|
| 1     | 13 | 20 | 35 | 14 | 15 | 26 | 18 | 27 | 16 | 19 |
| 2     | -7 | -4 | -9 | -5 | -6 | -7 | -9 | -8 | -7 | -9 |
| 3     | 5  | 3  | 2  | 1  | 4  | 3  | 5  | 2  | 1  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------|----|----|----|----|----|----|----|----|----|----|
| 1     | 21 | 24 | 28 | 15 | 16 | 18 | 24 | 14 | 12 | 28 |
| 2     | -6 | -7 | -9 | -8 | -7 | -9 | -5 | -7 | -9 | -7 |
| 3     | 3  | 2  | 1  | 5  | 2  | 3  | 1  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |    |    |





# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 | 16 | 11 | 12 | 13 |
| 2     | -5 | -6 | -7 | -8 | -9 | -4 | -5 | -6 |
| 3     | 4  | 3  | 2  | 1  | 3  | 5  | 2  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 14 | 15 | 16 | 11 | 12 | 13 | 14 | 15 |
| 2     | -7 | -8 | -9 | -4 | -5 | -6 | -7 | -8 |
| 3     | 3  | 2  | 4  | 3  | 5  | 1  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

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| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 16 | 11 | 12 | 13 |
| 2     | -9 | -4 | -5 | -6 |
| 3     | 2  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |

15 – 3 =

14 – 4 =

13 – 5 =

12 – 6 =

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 14 | 16 | 11 | 13 | 15 | 17 | 10 |
| 2     | -6 | -8 | -9 | -5 | -7 | -9 | -8 | -4 |
| 3     | 3  | 2  | 4  | 3  | 2  | 4  | 3  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 16 | 14 | 12 | 10 | 11 | 13 | 15 |
| 2     | -9 | -8 | -7 | -6 | -5 | -4 | -7 | -9 |
| 3     | 2  | 4  | 3  | 5  | 2  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 17 | 15 | 13 | 11 |
| 2     | -9 | -7 | -5 | -4 |
| 3     | 3  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |



$16 - 7 =$



$15 - 8 =$



$14 - 9 =$



$13 - 4 =$

# Abacus Calculation: Decomposition of 10 (-4)



-4 = -10 + 6    Fingering Ex.: 10-4, 11-4, 12-4

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 2     | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 |
| 3     | 3  | 5  | 2  | 4  | 3  | 1  | 5  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 2     | -4 | -4 | -4 | -4 | -4 | -4 | -4 | -4 |
| 3     | 4  | 5  | 3  | 2  | 1  | 5  | 3  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 10 | 12 | 14 | 16 |
| 2     | -4 | -4 | -4 | -4 |
| 3     | 4  | 3  | 2  | 1  |
| Ans.  |    |    |    |    |

10 - 4 =

11 - 4 =

12 - 4 =

13 - 4  
= 13-10+6  
= 9

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 24 | 15 | 28 | 11 | 19 | 22 | 17 | 14 |
| 2     | -9 | 4  | -7 | 5  | -6 | -8 | 5  | 3  |
| 3     | 3  | -6 | 5  | -9 | 3  | 4  | -3 | -5 |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 26 | 18 | 15 | 20 | 13 | 21 | 16 | 22 |
| 2     | -9 | -8 | 4  | -7 | 6  | -9 | 3  | -7 |
| 3     | 4  | 3  | -2 | 5  | -4 | 2  | -5 | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 28 | 16 | 24 | 18 |
| 2     | -9 | 5  | -8 | -6 |
| 3     | 3  | -7 | 2  | 3  |
| Ans.  |    |    |    |    |

14 – 5 =

15 – 6 =

13 – 4 =

12 – 3 =

## Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 26 | 14 | 18 | 22 | 16 | 28 | 15 | 19 |
| 2     | -9 | 5  | -7 | 5  | 4  | -9 | 3  | -6 |
| 3     | 3  | -6 | 4  | -3 | -7 | 5  | -4 | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 24 | 12 | 26 | 15 | 18 | 21 | 23 | 14 |
| 2     | -8 | 4  | -9 | 3  | 4  | -5 | -9 | 5  |
| 3     | 3  | -5 | 2  | -6 | -7 | 3  | 4  | -2 |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 20 | 26 | 12 | 28 |
| 2     | -8 | -7 | 5  | -9 |
| 3     | 5  | 3  | -4 | 1  |
| Ans.  |    |    |    |    |



## Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 26 | 18 | 24 | 15 | 28 | 17 | 22 | 13 |
| 2     | -9 | 5  | -7 | 4  | -9 | 5  | -8 | 4  |
| 3     | 3  | -6 | 4  | -3 | 2  | -4 | 3  | -6 |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 20 | 14 | 26 | 16 | 24 | 18 | 15 | 27 |
| 2     | -7 | 5  | -9 | 3  | -8 | -6 | 4  | -9 |
| 3     | 3  | -2 | 4  | -5 | 2  | 3  | -7 | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 19 | 23 | 14 | 26 |
| 2     | -8 | 5  | -3 | -9 |
| 3     | 3  | -7 | 5  | 3  |
| Ans.  |    |    |    |    |



# Match the Mascot to its House

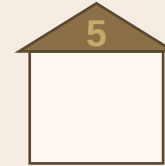
Each mascot's problem solves to a house number



$$14 - 6$$



$$13 - 5$$



$$12 - 4$$



$$11 - 3$$



$$15 - 7$$



$$16 - 8$$



$$17 - 9$$



$$10 - 2$$



## Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 24 | 31 | 16 | 19 | 27 | 14 | 28 |
| 2     | -5 | -9 | -7 | 4  | 5  | -9 | 3  | -8 |
| 3     | 3  | 4  | 5  | -6 | -3 | 4  | -5 | 2  |
| Ans.  |    |    |    |    |    |    |    |    |



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 24 | 35 | 15 | 16 | 26 | 18 | 27 |
| 2     | -5 | -9 | -8 | 4  | 3  | -7 | -9 | -8 |
| 3     | 4  | 3  | 2  | -6 | -5 | 3  | 5  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |



| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 21 | 28 | 24 | 15 |
| 2     | -6 | -7 | -9 | -8 |
| 3     | 3  | 4  | 1  | 5  |
| Ans.  |    |    |    |    |





# Count and Subtract

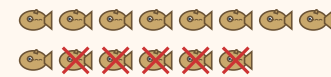
Cross out the smaller group and write the difference



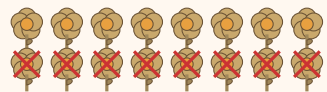
$$12 - 4 = \boxed{\phantom{00}}$$



$$15 - 7 = \boxed{\phantom{00}}$$



$$14 - 5 = \boxed{\phantom{00}}$$



$$16 - 8 = \boxed{\phantom{00}}$$



$$13 - 6 = \boxed{\phantom{00}}$$



$$11 - 2 = \boxed{\phantom{00}}$$



# Abacus Calculation: Combination of 10 (–9 two-step)



–9 = –10 + 5 – 4    Fingering Ex.: 14–9

| S.No.  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|--------|----|----|----|----|----|----|----|----|
| 1      | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 14 |
| 2      | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3      | 3  | 5  | 2  | 4  | 1  | 3  | 5  | 2  |
| * Ans. |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 14 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 4  | 3  | 2  | 5  | 1  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 14 | 14 | 14 | 14 |
| 2     | -9 | -9 | -9 | -9 |
| 3     | 5  | 4  | 2  | 3  |
| Ans.  |    |    |    |    |

14 – 9 =

15 – 9 =

16 – 9 =

14 – 9  
= 14-10+5-4  
= 5

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 14 | 15 | 16 | 14 | 15 | 16 | 14 | 15 |
| 2     | -9 | -9 | -9 | -9 | -9 | -9 | -9 | -9 |
| 3     | 3  | 2  | 4  | 1  | 3  | 5  | 2  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 14 | 12 | 13 | 14 | 15 | 16 | 14 |
| 2     | -8 | -8 | -7 | -7 | -8 | -9 | -9 | -9 |
| 3     | 2  | 3  | 4  | 3  | 1  | 5  | 3  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 15 | 14 | 16 | 13 |
| 2     | -9 | -9 | -9 | -8 |
| 3     | 1  | 3  | 2  | 4  |
| Ans.  |    |    |    |    |



$13 - 8 =$



$12 - 7 =$



$14 - 8 =$



$15 - 8 =$

# Abacus Calculation: Combination of 10 (–8 two-step)



–8 = –10 + 5 – 3    Fingering Ex.: 13–8, 14–8

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 14 | 13 | 14 | 13 | 14 | 13 | 14 |
| 2     | -8 | -8 | -8 | -8 | -8 | -8 | -8 | -8 |
| 3     | 4  | 3  | 5  | 2  | 1  | 4  | 3  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 14 | 13 | 14 | 13 | 14 | 13 | 14 | 13 |
| 2     | -8 | -8 | -8 | -8 | -8 | -8 | -8 | -8 |
| 3     | 2  | 4  | 1  | 3  | 5  | 2  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 13 | 14 | 13 | 14 |
| 2     | -8 | -8 | -8 | -8 |
| 3     | 2  | 5  | 4  | 1  |
| Ans.  |    |    |    |    |

13 – 8 =

14 – 8 =

15 – 8 =

13 – 8  
= 13-10+5-3  
= 5

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 14 | 15 | 12 | 13 | 14 | 15 | 16 |
| 2     | -7 | -8 | -9 | -7 | -8 | -9 | -7 | -8 |
| 3     | 3  | 2  | 4  | 3  | 5  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 | 16 | 13 | 14 | 15 |
| 2     | -7 | -8 | -9 | -7 | -8 | -9 | -7 | -8 |
| 3     | 5  | 3  | 2  | 4  | 3  | 5  | 2  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 16 | 12 | 14 | 15 |
| 2     | -9 | -7 | -8 | -9 |
| 3     | 4  | 1  | 5  | 3  |
| Ans.  |    |    |    |    |

$13 - 7 =$

$14 - 7 =$

$15 - 7 =$

$16 - 7 =$

# Abacus Calculation: Combination of 10 (-7 two-step)



-7 = -10 + 5 - 2    Fingering Ex.: 12-7, 13-7

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 12 | 13 | 12 | 13 | 12 | 13 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 3  | 5  | 4  | 2  | 1  | 4  | 5  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 12 | 13 | 12 | 13 | 12 | 13 | 12 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 2  | 4  | 1  | 3  | 5  | 2  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 12 | 13 | 12 | 13 |
| 2     | -7 | -7 | -7 | -7 |
| 3     | 2  | 5  | 4  | 1  |
| Ans.  |    |    |    |    |

12 - 7 =

13 - 7 =

14 - 7 =

12 - 7  
= 12-10+5-2  
= 5

# Abacus Calculation: Combination of 10 (-7 cont.)



-7 = -10 + 5 - 2    Fingering Ex.: 12-7, 13-7

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 12 | 13 | 12 | 13 | 12 | 13 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 5  | 3  | 2  | 4  | 3  | 5  | 1  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 12 | 13 | 12 | 13 | 12 | 13 | 12 |
| 2     | -7 | -7 | -7 | -7 | -7 | -7 | -7 | -7 |
| 3     | 4  | 3  | 1  | 4  | 2  | 5  | 3  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 12 | 13 | 12 | 13 |
| 2     | -7 | -7 | -7 | -7 |
| 3     | 1  | 4  | 5  | 3  |
| Ans.  |    |    |    |    |

12 - 7 =

13 - 7 =

15 - 7 =

16 - 7 =

Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 11 | 12 | 13 | 14 | 12 | 13 | 14 |
| 2     | -6 | -6 | -7 | -8 | -9 | -7 | -8 | -9 |
| 3     | 4  | 3  | 2  | 5  | 1  | 3  | 4  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 12 | 13 | 14 | 11 | 12 | 13 | 14 |
| 2     | -6 | -7 | -8 | -9 | -6 | -7 | -8 | -9 |
| 3     | 5  | 4  | 3  | 2  | 1  | 5  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 |
| 2     | -6 | -7 | -8 | -9 |
| 3     | 2  | 4  | 3  | 2  |
| Ans.  |    |    |    |    |



11 – 6 =



12 – 6 =



13 – 6 =



15 – 6 =



# Abacus Calculation: Combination of 10 (-6 two-step)



-6 = -10 + 5 - 1    Fingering Ex.: 11-6, 12-6

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 12 | 11 | 12 | 11 | 12 | 11 | 12 |
| 2     | -6 | -6 | -6 | -6 | -6 | -6 | -6 | -6 |
| 3     | 3  | 5  | 4  | 2  | 1  | 4  | 5  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 11 | 12 | 11 | 12 | 11 | 12 | 11 |
| 2     | -6 | -6 | -6 | -6 | -6 | -6 | -6 | -6 |
| 3     | 2  | 4  | 1  | 3  | 5  | 2  | 4  | 3  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 11 | 12 | 11 | 12 |
| 2     | -6 | -6 | -6 | -6 |
| 3     | 2  | 5  | 4  | 1  |
| Ans.  |    |    |    |    |

11 - 6 =

12 - 6 =

13 - 6 =

11 - 6  
= 11-10+5-1  
= 5

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 12 | 13 | 14 | 15 | 11 | 12 | 13 | 14 |
| 2     | -6 | -7 | -8 | -9 | -5 | -6 | -7 | -8 |
| 3     | 4  | 5  | 3  | 2  | 4  | 3  | 2  | 5  |
| Ans.  |    |    |    |    |    |    |    |    |



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 15 | 11 | 12 | 13 | 14 | 15 | 11 | 12 |
| 2     | -9 | -5 | -6 | -7 | -8 | -9 | -5 | -6 |
| 3     | 1  | 3  | 4  | 2  | 5  | 3  | 2  | 4  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 13 | 14 | 15 | 12 |
| 2     | -7 | -8 | -9 | -6 |
| 3     | 1  | 3  | 5  | 4  |
| Ans.  |    |    |    |    |



13 – 5 =

14 – 6 =

15 – 7 =

12 – 4 =

# Abacus Calculation: Revision



| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 13 | 12 | 11 | 10 | 15 | 14 | 13 | 12 |
| 2     | -9 | -8 | -7 | -6 | -9 | -8 | -7 | -6 |
| 3     | 3  | 4  | 2  | 5  | 3  | 2  | 4  | 1  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 11 | 10 | 15 | 14 | 13 | 12 | 11 | 10 |
| 2     | -9 | -8 | -7 | -6 | -9 | -8 | -7 | -6 |
| 3     | 5  | 3  | 2  | 4  | 3  | 5  | 4  | 2  |
| Ans.  |    |    |    |    |    |    |    |    |

| S.No. | 1  | 2  | 3  | 4  |
|-------|----|----|----|----|
| 1     | 15 | 14 | 13 | 12 |
| 2     | -9 | -8 | -7 | -6 |
| 3     | 1  | 3  | 4  | 5  |
| Ans.  |    |    |    |    |

11 – 7 =

12 – 8 =

13 – 9 =

14 – 5 =

# Mental Calculation

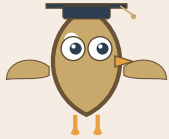


| S.No.    | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8 | 9  | 10 |
|----------|----|----|----|----|----|----|----|---|----|----|
| <b>A</b> | 2  | 5  | 7  | 4  | 6  | 3  | 8  | 2 | 5  | 7  |
| 2        | 3  | -3 | -3 | 2  | -2 | 5  | -4 | 3 | -2 | -5 |
| 3        | -1 | 4  | 2  | -3 | 3  | -2 | 2  | 4 | 1  | 3  |
| Ans.     |    |    |    |    |    |    |    |   |    |    |
| Ans.     |    |    |    |    |    |    |    |   |    |    |



## Two-Digit Addition: Carrying Forward

Now let's apply the trick to bigger numbers



Have you learned how to  
carry-forward a digit?  
Now do it for double digits!

100

|   |    |
|---|----|
| 1 | 99 |
| 2 | 98 |
| 3 | 97 |
| 4 | 96 |
| 5 | 95 |
| 6 | 94 |
| 7 | 93 |
| 8 | 92 |
| 9 | 91 |

Two-digit examples:

$$\begin{array}{r} 29 \\ + 61 \\ \hline 90 \end{array}$$

$$\begin{array}{r} 47 \\ + 53 \\ \hline 100 \end{array}$$

$$\begin{array}{r} 38 \\ + 42 \\ \hline 80 \end{array}$$

$$\begin{array}{r} 65 \\ + 35 \\ \hline 100 \end{array}$$



# Mental Calculation



| S.No.    | 1  | 2  | 3 | 4  | 5  | 6 | 7  | 8  | 9  | 10 |
|----------|----|----|---|----|----|---|----|----|----|----|
| <b>A</b> | 3  | 7  | 2 | 8  | 5  | 2 | 8  | 4  | 7  | 5  |
| 2        | 5  | -5 | 4 | -6 | -2 | 3 | -6 | 2  | -4 | 3  |
| 3        | -2 | 2  | 3 | 2  | 4  | 5 | 1  | -3 | 1  | -2 |
| Ans.     |    |    |   |    |    |   |    |    |    |    |
| Ans.     |    |    |   |    |    |   |    |    |    |    |

| S.No.    | 1 | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 |
|----------|---|----|----|----|----|----|---|----|----|----|
| <b>B</b> | 2 | 7  | 3  | 6  | 2  | 8  | 5 | 4  | 6  | 5  |
| 2        | 3 | -5 | 4  | 3  | 7  | -3 | 2 | -3 | 2  | 3  |
| 3        | 2 | 1  | -2 | -5 | -5 | 2  | 1 | 5  | -4 | -2 |
| Ans.     |   |    |    |    |    |    |   |    |    |    |
| Ans.     |   |    |    |    |    |    |   |    |    |    |

## Listening Exercise

1 \_\_\_\_\_

2 \_\_\_\_\_

3 \_\_\_\_\_

4 \_\_\_\_\_

5 \_\_\_\_\_

6 \_\_\_\_\_

7 \_\_\_\_\_

8 \_\_\_\_\_

9 \_\_\_\_\_

10 \_\_\_\_\_

# Final Assessment

Sections B, C, D — show what you have learned



Time: \_\_\_\_\_

## Section B — Abacus Calculation

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|-------|----|----|----|----|----|----|----|----|
| 1     | 18 | 21 | 30 | 28 | 17 | 25 | 16 | 12 |
| 2     | -5 | 4  | -8 | -4 | 6  | 4  | 5  | 9  |
| 3     | -4 | 3  | -9 | 6  | -7 | 2  | 8  | -6 |
| Ans.  |    |    |    |    |    |    |    |    |

## Section C — Mental Calculation

| S.No. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------|----|----|----|----|----|----|----|----|----|----|
| 1     | 9  | 7  | 4  | 6  | 5  | 8  | 2  | 9  | 3  | 6  |
| 2     | -3 | -3 | 5  | 2  | -2 | 2  | 7  | -7 | 9  | -5 |
| 3     | 2  | 5  | -2 | -1 | 8  | -7 | -3 | 5  | -2 | 9  |
| Ans.  |    |    |    |    |    |    |    |    |    |    |

## Section D — Fill in the Combination & Decomposition Number

